NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Pediatric Considerations in Cardiology Part 2

Principles and Practices of Osteopathic Medicine II

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

1. Students will be able to recognize signs and symptoms suggestive of cardiac issues in infants
2. Student will be able to discuss factors which suggest an innocent murmur vs a murmur that needs further evaluation
3. Student will be able to identify clinical, anatomical, and physiologic aspects of atrial septal defects (ASD) and ventricular septal defects (VSD)
4. Student will be able to identify clinical, anatomical, and physiologic aspects of other acyanotic congenital heart lesions including A-V Canal, Patent ductus arteriosus, and Coarctation of the aorta
5. Student will be able to identify clinical aspects of specific cardiac arrhythmias- Supraventricular tachycardia, Prolonged Q-T syndrome, and Congenital heart block in infants

Infants with possible cardiac problem

History

- Prenatal Hx
- Feeding difficulty- Prolonged
- Sweating with feeding
- Irritability
- Color change- pale or blue

Physical

- Poor weight gain/ Failure to thrive
- Congenital defects/ Malformations
- Tachypnea
- Tachycardia/ Bradycardia
- Murmurs
- Poor perfusion
- Central cyanosis
- Pale/grey/ poor perfusion
- Absent/Delayed femoral pulse
- Quality of pulses
- Respiratory issues- infections/ wheezing
- Hepatomegaly

Principles…

Blood flows from high pressure to low pressure	Look, Listen, Feel	They are called Vitals for a reason
Growth charts matter in Pediatrics	Murmurs don't always mean cardiac Dz	No Murmur doesn't always mean No Cardiac Dz

Evaluation of Murmurs in Infants/Children

- Grade 1-6
- Where? Don't forget the back in infants
- What? Holosystolic, systolic, diastolic, clicks, gallops (S3,S4)
- Position? Sitting, supine, left lateral decubital

Innocent Murmurs

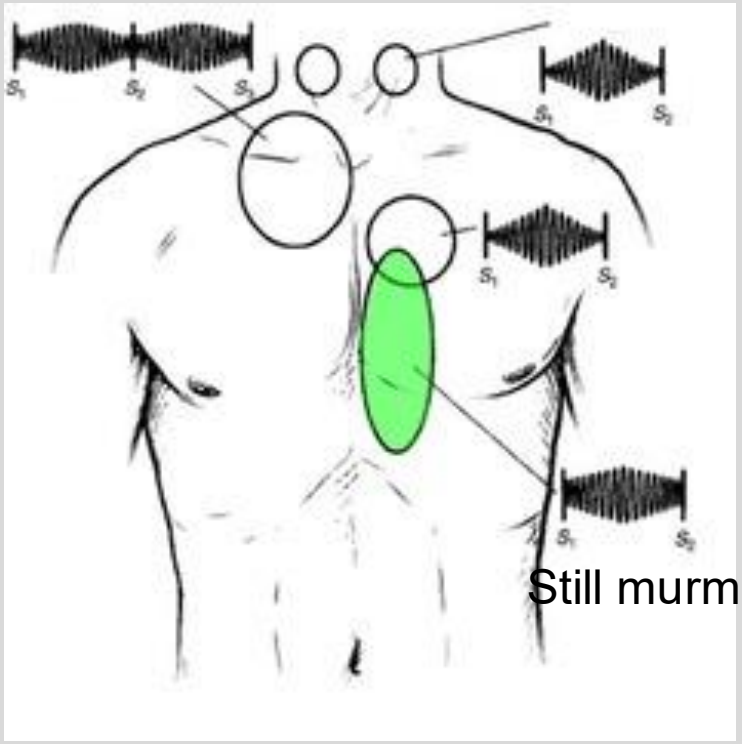
- Very….very Common
- Benign history and physical
- Fever, Anemia may exacerbate murmur
- Reassurance

Common Innocent Murmurs

- Still Murmur
- Venous Hum
- Peripheral/ Branch Pulmonic Stenosis

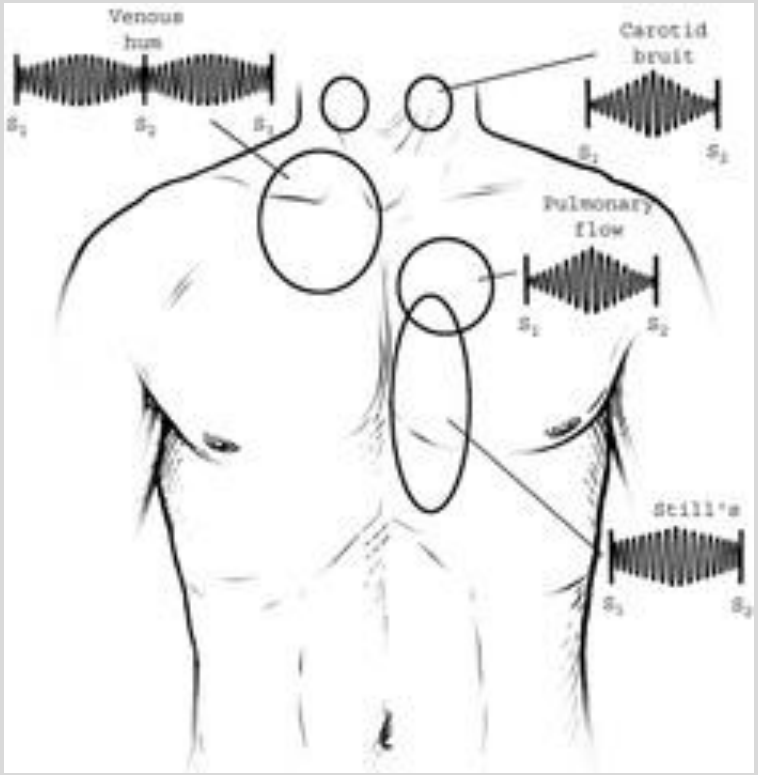
Stills Murmur

- Musical/ Vibratory
- Low pitch SEM
- Grade 1-3
- Increases while supine/ lying down and high flow states



Venous Hum

- From venous return
- Medium pitch/ Blowing
- Continuous murmur
- Upper Rt sternal border/ infraclavicular area
- **Disappears when supine or turning neck**



Murmurs suggesting further evaluation

- Loud (grade 4-6), Harsh qualities, thrills
- History suggesting a cardiac issue, pul issues, constitutional complaints
- Other Physical- eg, poor weight gain, doesn't look well, abnormal vitals

Acyanotic Heart disease

Volume load- Left to Rt shunts

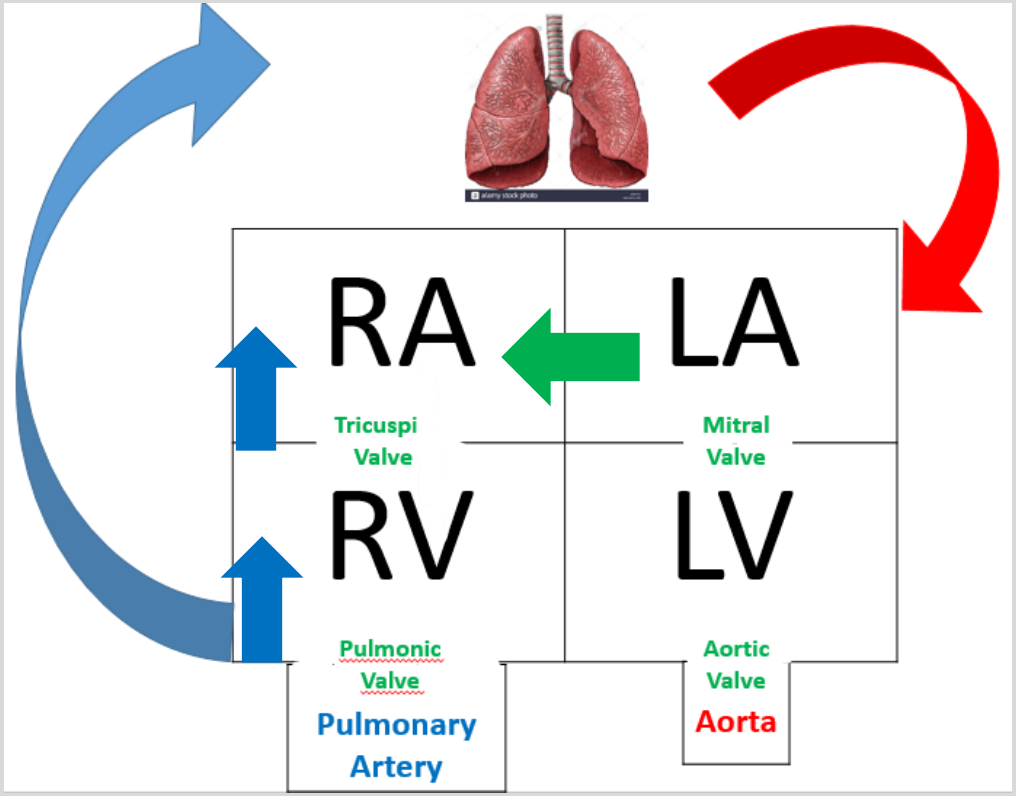
- ASD
- VSD
- PDA
- A-V canal/ Endocardial cushion defect

Obstructive Lesions-pressure load

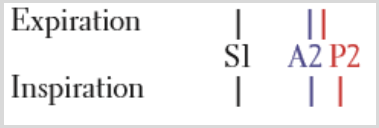
- Coarctation of the aorta
- Valvular Pulmonic Stenosis
- Valvular Aortic stenosis

Atrial Septal Defect (ASD)

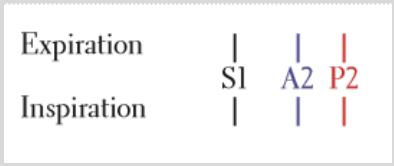
- Ostium Secundum defects: most common
- **L → R** Shunt
- Murmur:
  - **Loud S1, fixed split S2 (no change with resp)**
- Assoc Fetal Alcohol Syndrome



Normal



ASD



First Aid for the USMLE

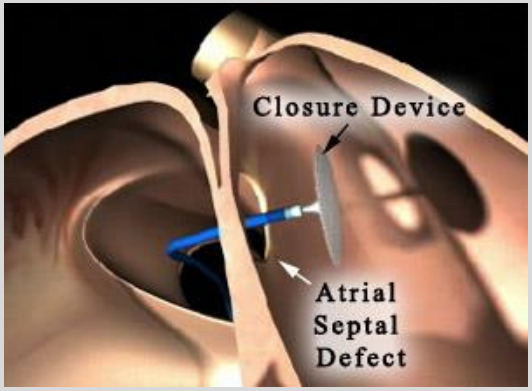
ASD

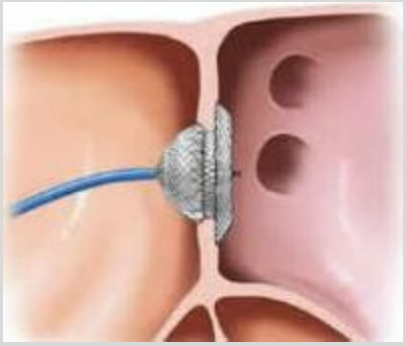
Diagnosis

Usually asymptomatic, hear murmur

- Echocardiogram

Many go undiagnosed and cause issues much later in life

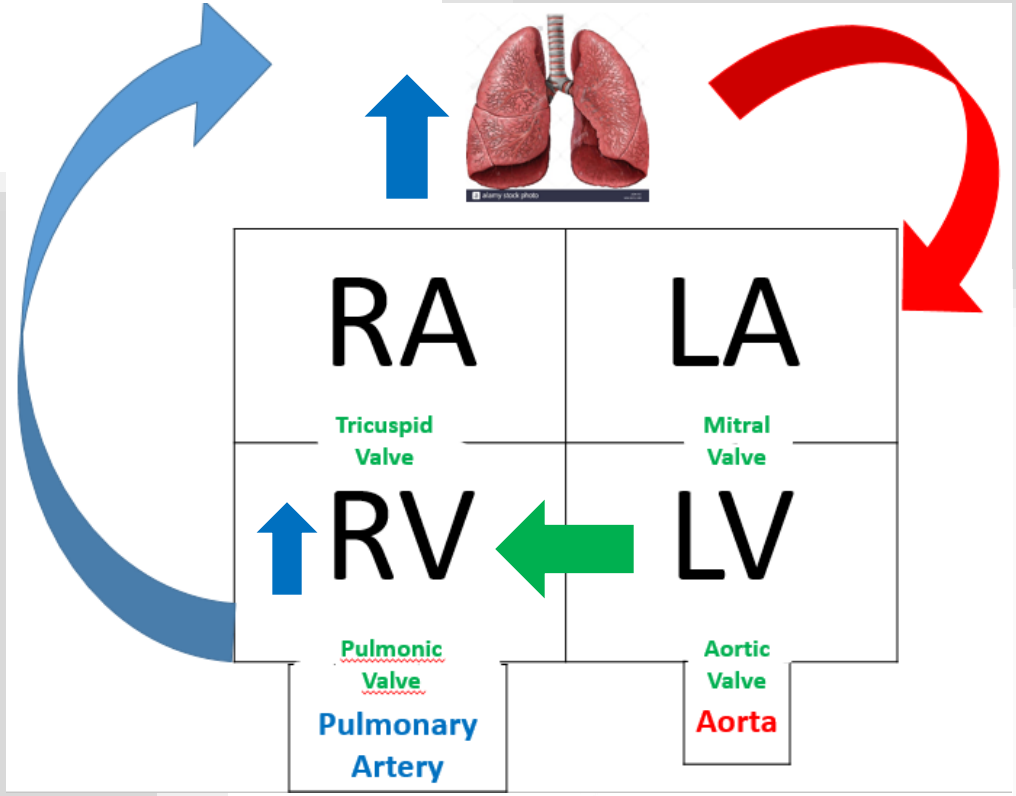


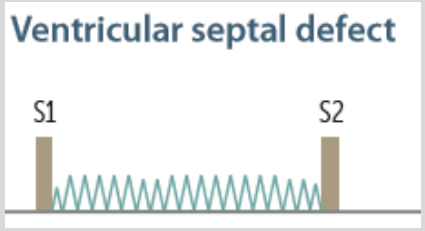


- Treatment:
- Watch and wait- spontaneous closures (small and med)
- Antibiotic prophylaxis NOT usually recommended
  - Surgical
  - Transcatheter device

Ventricular Septal Defect (VSD)

- Most common
- **L → R** Shunt
- Murmur
  - Holosystolic
- **Assoc with Down Syndrome**



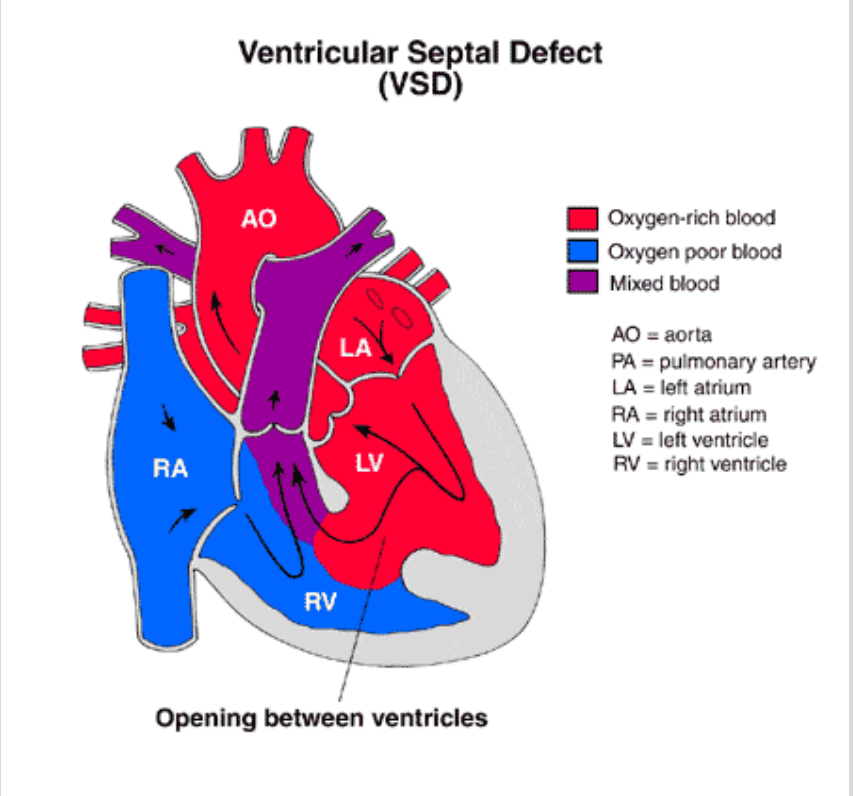


First Aid for the USMLE

Ventricular Septal Defect (VSD)

- Symptoms depend on size of VSD and Pul pressures/ blood flow
- Holosystolic murmur- harsh, blowing, left lower sternal border, thrill

  (May not have murmur in newborn period)
- Resp, growth problems, heart failure if mod/large VSD
- Risk: **Eisenmenger Syndrome over long term**



VSD

Small VSD

- Most small and muscular ones close spontaneously
- Reassurance/ monitor clinically in office and regular f/u with cardio
- No restrictions in activity
- No ABX prophylaxis for dental/ surgical procedures

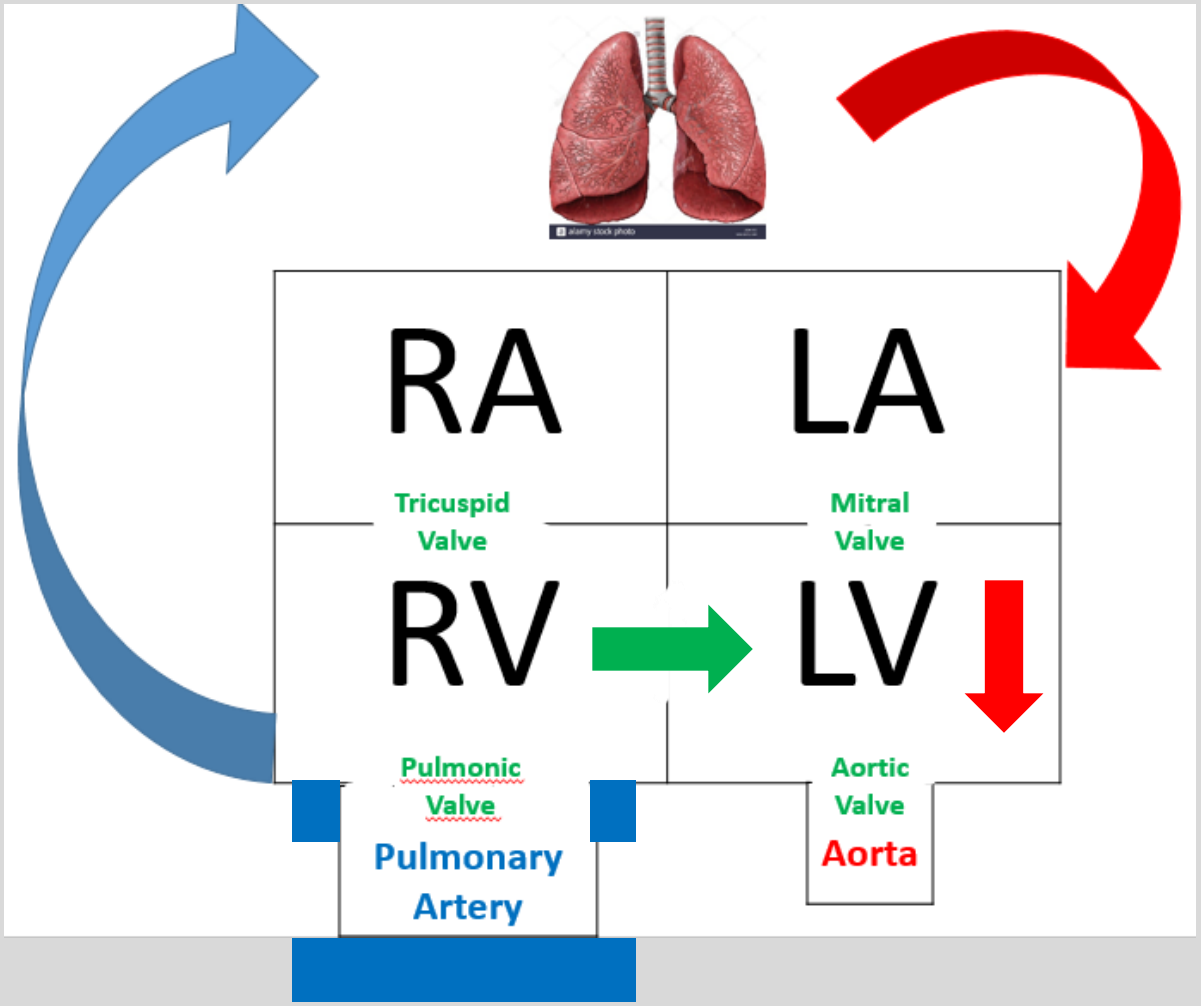
Large VSD

- Control heart failure and prevent pul vasc Dz
- Tx Medically or surgically
- Surgical repair excellent prognosis

Eisenmenger Syndrome

- Result of uncorrected L → R shunt over time
- Volume overload of R heart → increase R sided pressure
- Shunt **REVERSAL** from L → R to **R→ L**
- Deoxygenated blood spilling into general circulation → step down in O2 → **cyanosis**

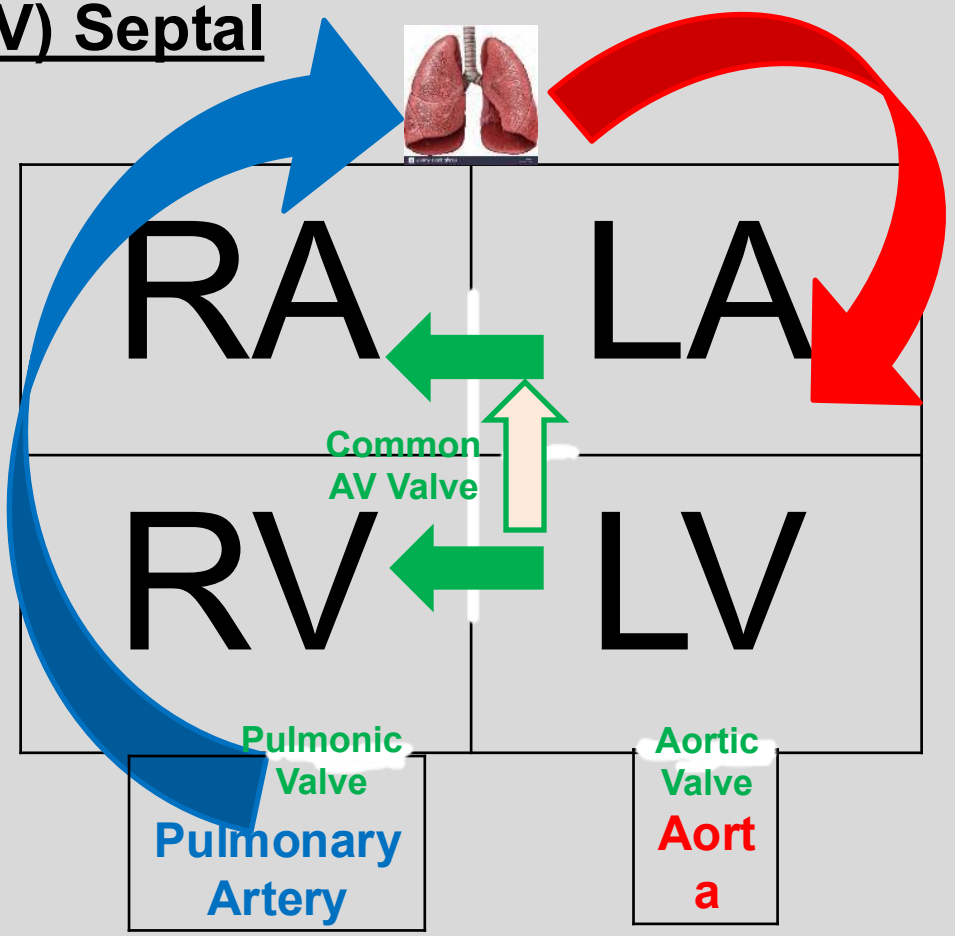




International Journal of Cardiology, Volume 177, Issue 1

Complete Atrioventricular (AV) Septal Defect

- VSD, ASD, & Common AV valve
- L → R shunt & severe AV regurgitation
- Result: Volume Overload → excess pulmonary blood flow
- CXR: **increased pulmonary markings & cardiomegaly**
- **Assoc Down syndrome**

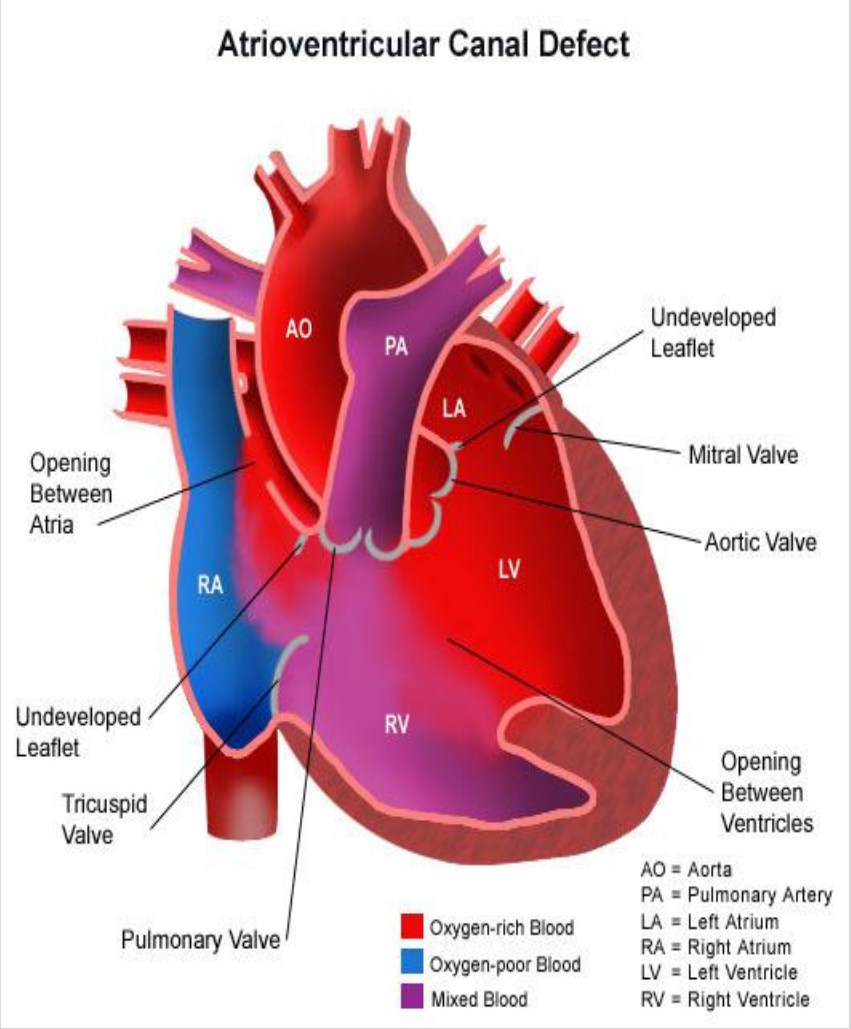




A-V canal/ Endocardial cushion defect

- Murmur: ASD/VSD
- Rt sided overload
- Eventual pul HTN leads to Rt to Lt shunting (cyanosis)
  - **Eisenmenger Syndrome**

Surgical Repair

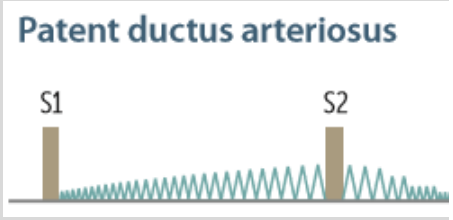


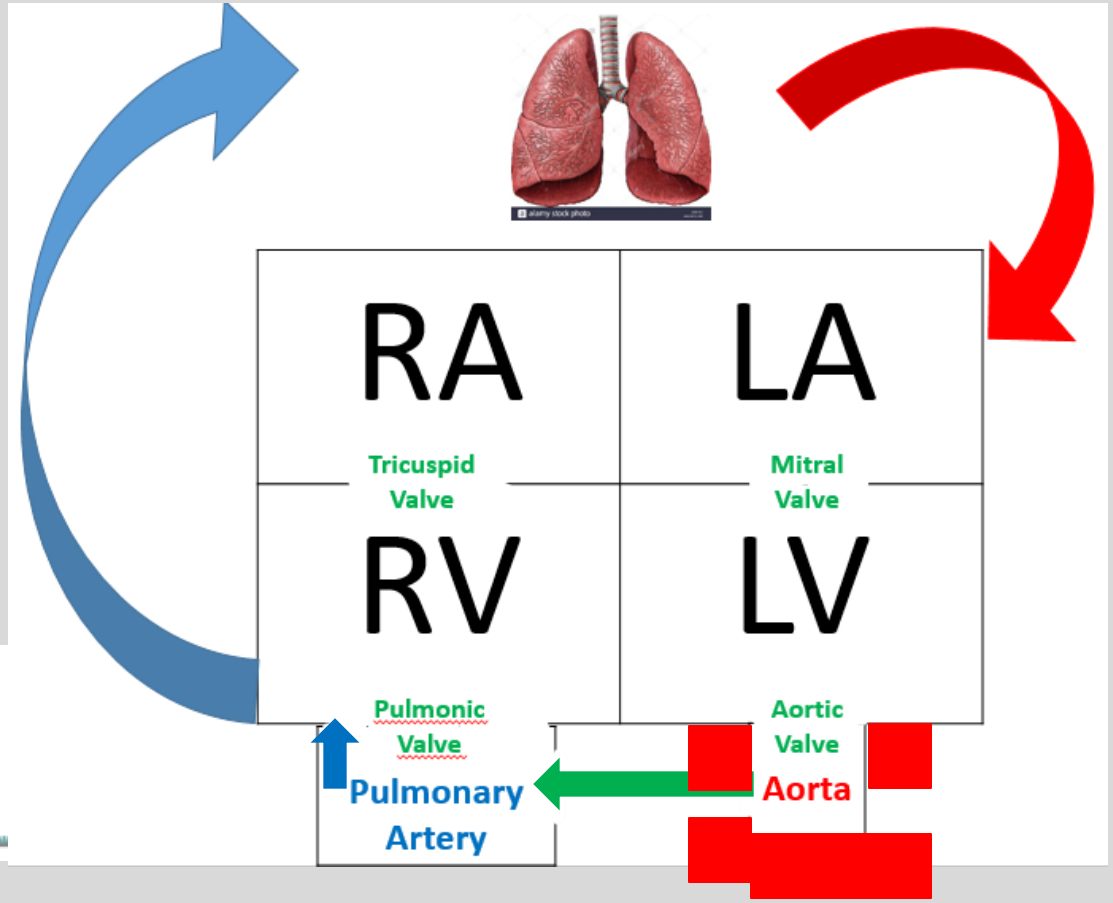
Patent Ductus Arteriosus (PDA)

- Persistent connection between aorta and pulmonary artery
- L → R shunt:
- Murmur
  - **Machine like continuous**
  - Thrill 2nd left interspace- radiation down

- **Assoc Congenital Rubella**

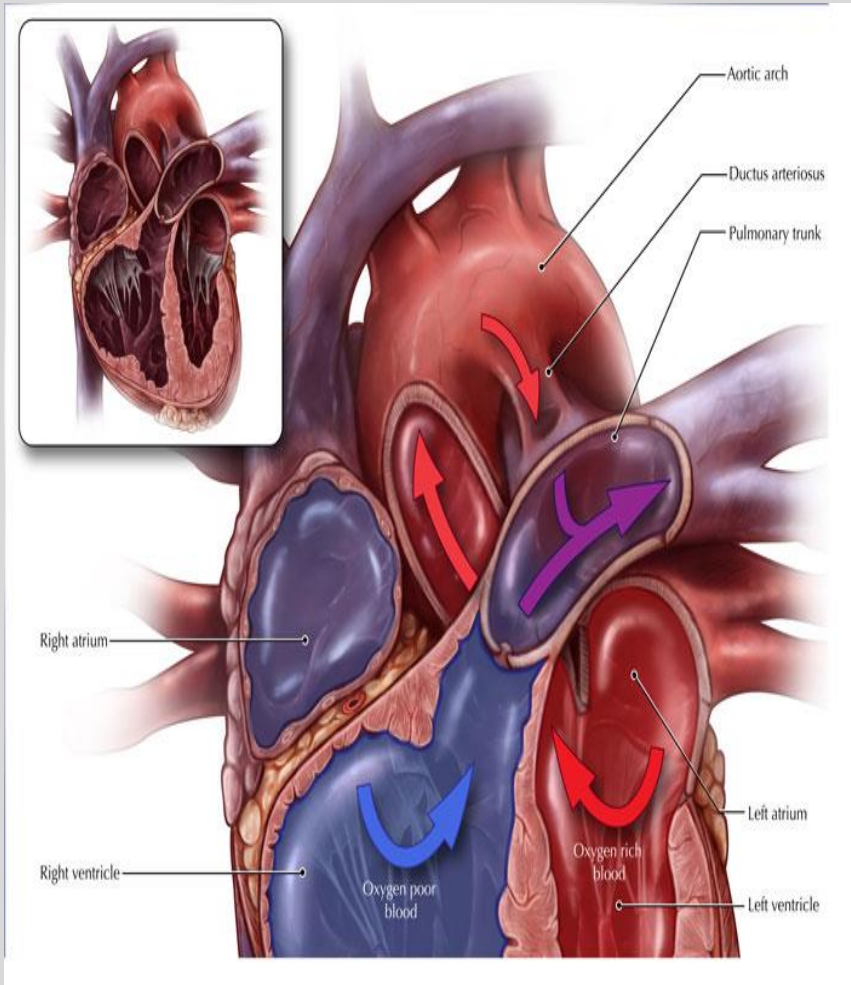






Netter's Infectious Diseases
First Aid for the USMLE

Patent Ductus Arteriosus

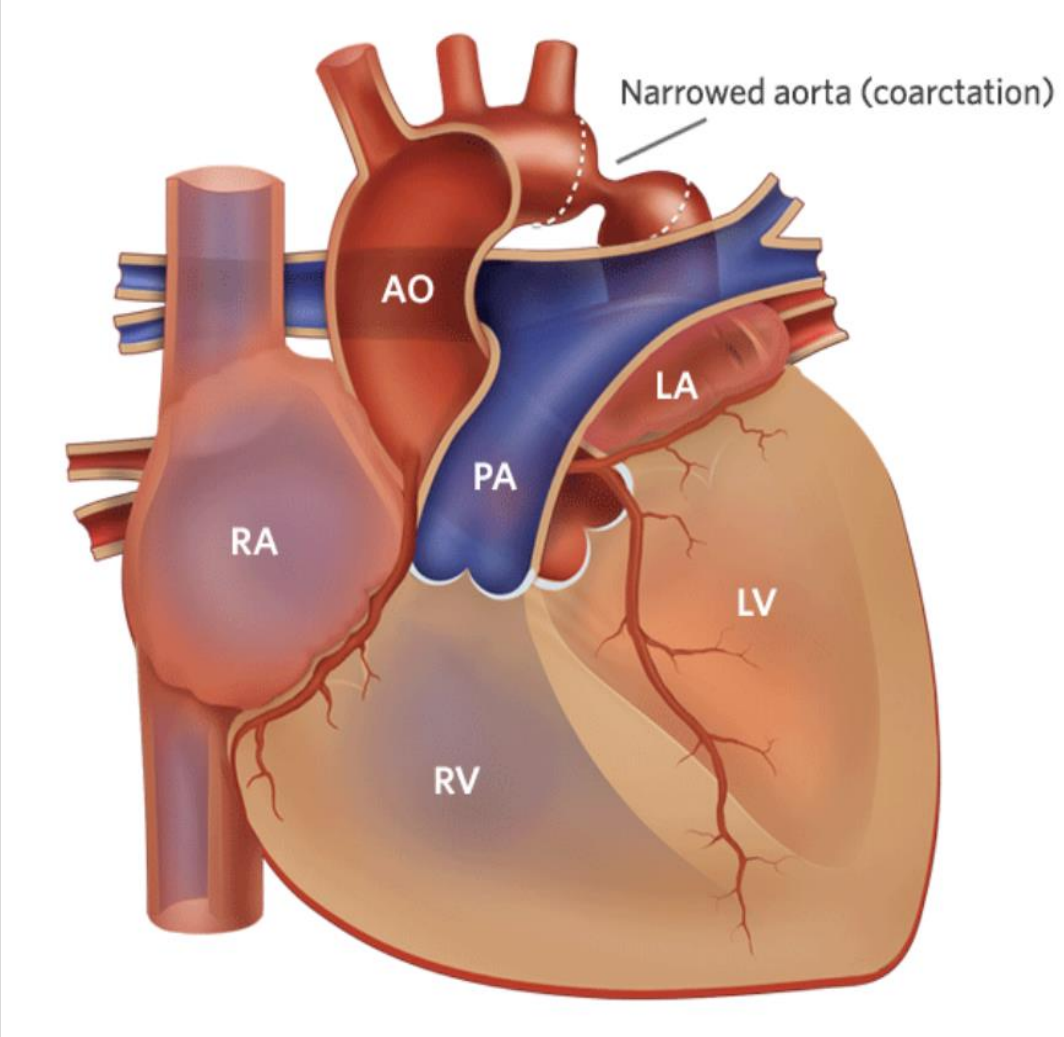


- Premies more common- may close on own
- Term infant with PDA unlikely to close on own
- Large acts like VSD (L to R shunt) and cause shunt reversal over time
- Need to close ALL PDA's that remain open
- Treatment: **Indomethacin** (premies),
- Surgical correction- coils, ligation, umbrellas

Coarctation of the Aorta

- Narrowing of the aorta
  - Juxtaductal – near insertion of Ductus Arteriosus
  - Distal to the aortic arch
- Lower extremity…
  - **Claudication**
  - **Hypotension (lower BP legs)**
  - **Absent femoral pulse or Brachial-femoral delay**
  - **Cyanosis (legs)**
- **Upper extremity Hypertension**
  - **Berry Aneurysm**
  - **Heart Failure**

  Body Compensates
  - Collateral circulation
  - Intercostal arteries grow → erode ribs (**"rib notching"** on CXR)
- Bicuspid Aortic Valve, **Turner's Syndrome**



Perloff's Clinical Recognition of Congenital Heart Disease, 6th ed.

Coarctation of the Aorta

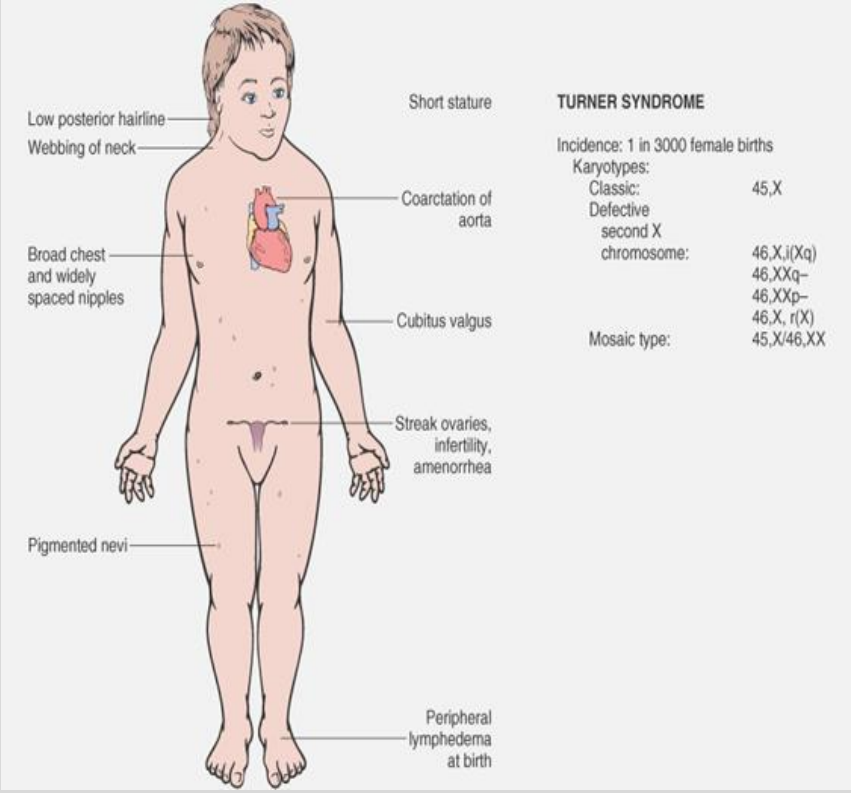
Diagnosis

- 4 limb Blood pressure and pulse ox
- Absent femoral pulse…
- …or Brachial Femoral delay (palpate together)
- Varying murmurs (murmur maybe in the back)
- Older patients
  - Hypertension
  - LVH
  - Rib notching on CXR
  - Stroke

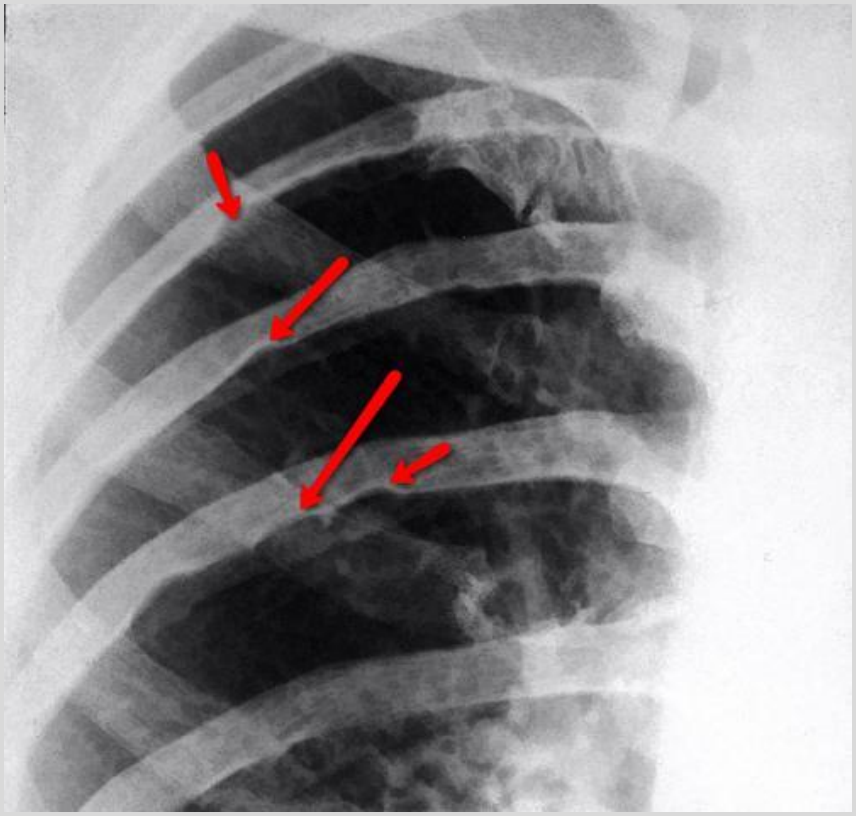


Coarctation Cont.…

Turner Syndrome XO



Rib Notching- later finding older child/adult



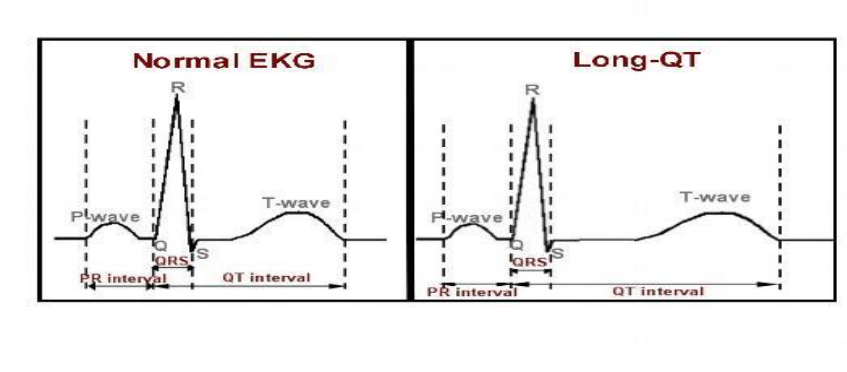
Electrical Problems

Get an EKG!!!!

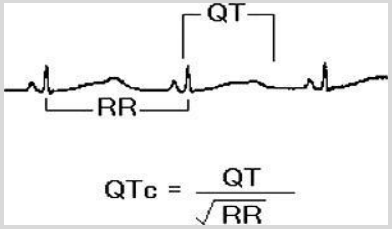
Long QT syndrome

Sudden Death/SIDS

- Genetic mutations
  - Romano Ward
  - Jervell-Lange-Nielsen (hearing loss)
- Family history

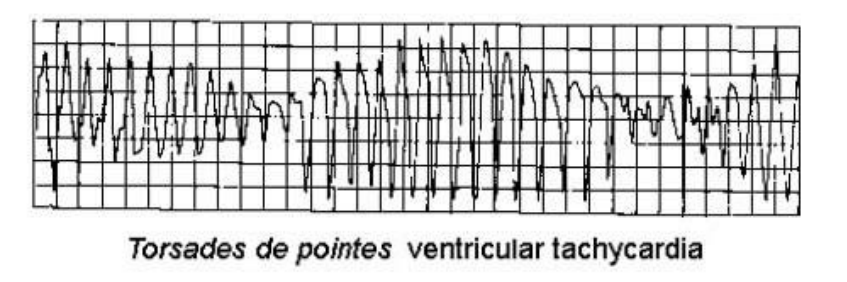


- Clinically: syncope (esp after a startle, sudden fright), or with exercise
- May present with: Palpitations, syncope, seizures
- **Medications may precipitate/ prolong QT**
- QTc (QT corrected)



Long QT syndrome

- Malignant arrhythmia
  - Torsades de pointes
  - Ventricular fibrillation



Torsades de pointes ventricular tachycardia

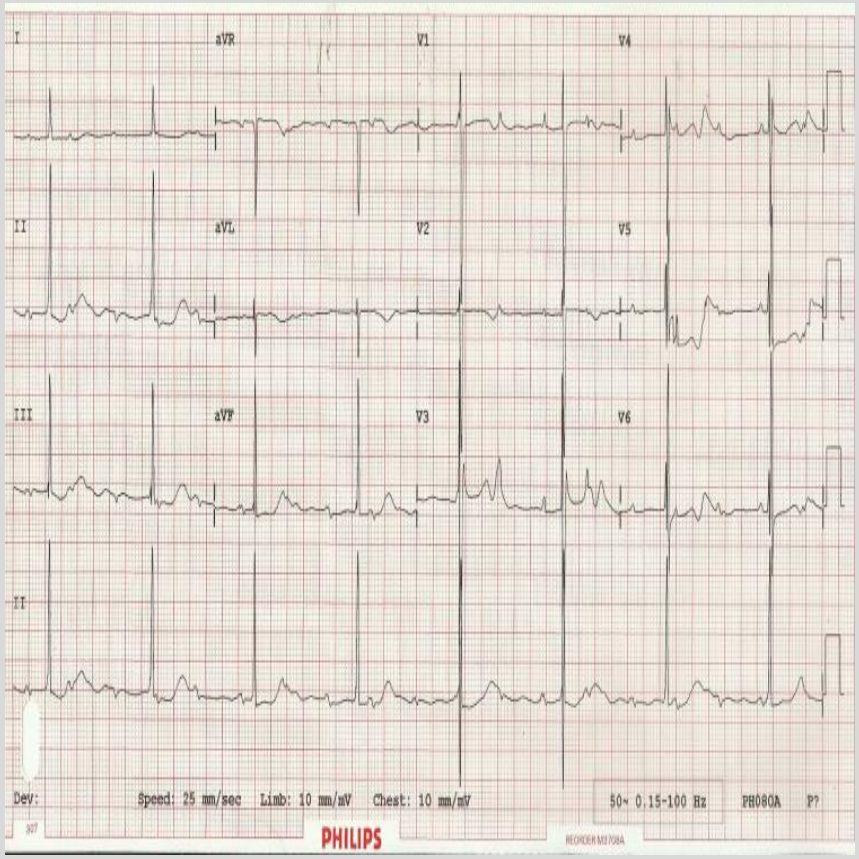
- Avoid certain medications if known Prolonged QT
- Any medication needs to be checked and cleared
- Cardio/ Genetics consult
- Beta blockers
- Implantable cardiac defibrillator

Congenital Heart Block- 3rd degree heart block
HR TOO SLOW!!!

- Maternal autoimmune disease
  - Systemic Lupus Erythematous(SLE)
  - Sjogren syndrome
- Autoimmune injury to fetal conduction system by IgG
- Anti-SSA/Ro, anti-SSB/La
- HR- very slow 60-80 beats…sometimes slower (ventricular rate)
- Infants may be asymptomatic
- Many detected intrauterine- fetal echo

Neonatal Lupus Syndrome

Congenital Heart block



Rash





http://www.jpgo.org/2014/05/congenital-heart-block-due-to-maternal.html

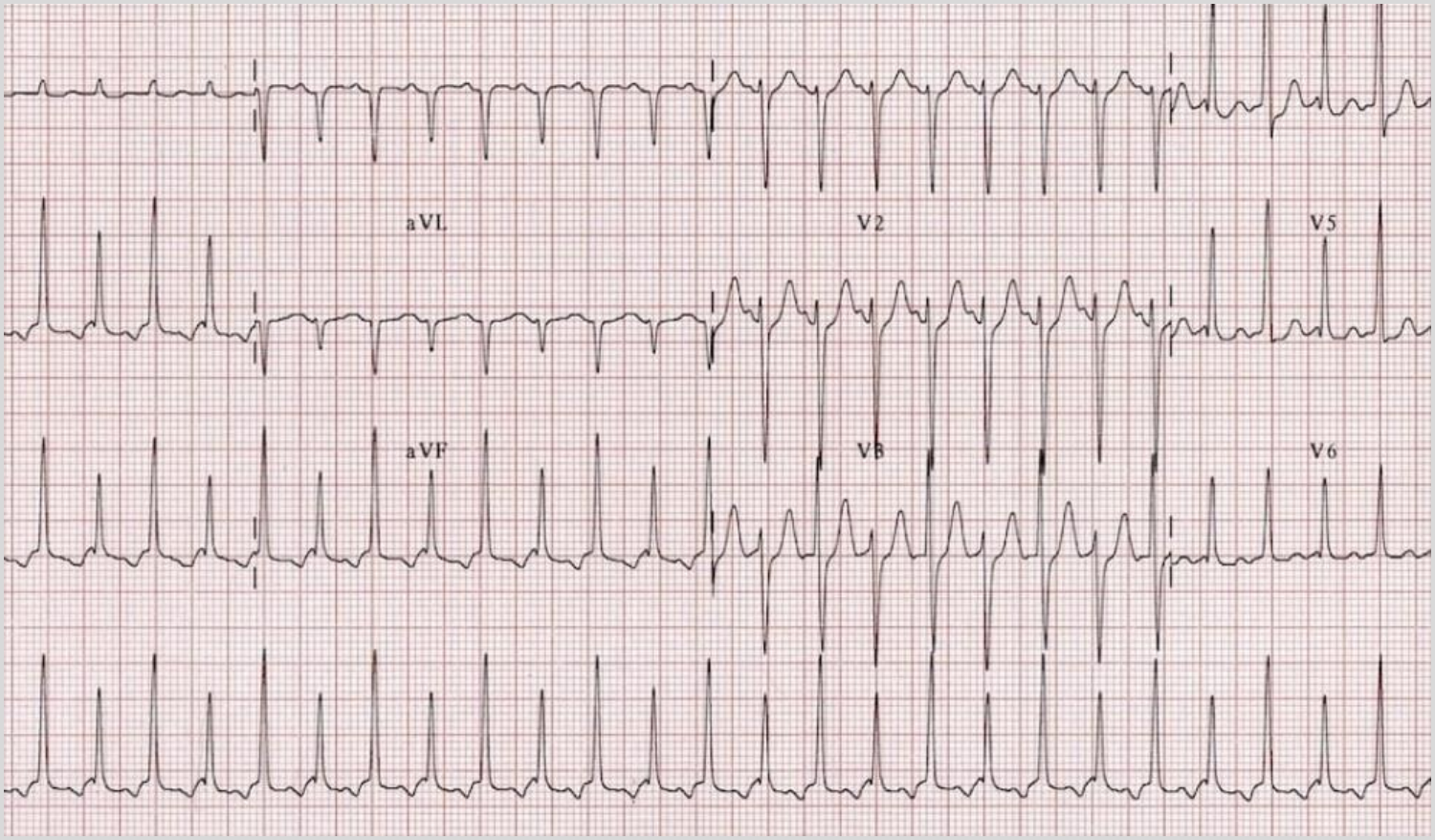
Supraventricular Tachycardia –SVT
HR TOO FAST!!!

- HR > 180 Bpm
- Abrupt onset and cessation
- Irritable infant, restless
- Narrow complex QRS
- No discernable P waves
- Wolf Parkinson White association

- Treatment
- Vagal maneuvers
  - Icebag to face- infants
  - Face in Ice water in older
  - Valsalva, carotid massage
- Stable Pts- Adenosine rapid IV push
- Unstable Pts- synchronized cardioversion



SVT



https://litfl.com/supraventricular-tachycardia-svt-ecg-library/

- A 3 year old girl present to the clinic for a well check. Asymptomatic, no complaints, dev appropriate. Growth charts are normal. Picky eater, drinks a lot of milk and not much else.
- Vitals normal
- You note a 2/6 SEM, heard best to left side of chest, tricuspid, erbs and mitral areas. Louder when lying down. NI femoral pulse. Chest clear, no HSM
- What is the most likely next course of action?
  - Echo
  - ECG
  - Give PGE1
  - Give Indomethecin
  - Reassure and follow

Question

- 2 week old male infant presents to the clinic for a follow up weight check. Parents say that he has been fussy and taking over an hour to finish a 3 oz bottle, he will stop every few minutes and seems to be sweating. They say his color is good and he has never turned blue. Weight today is the same as it was at DOL 4
- Q: are you worried about this infant? YES
- PE HR 128, Pox 92%, RR 42, Wt 7lb-2 oz, (BW 7lb-3oz)
- Alert, NAD
- Pale color, 4/6 murmur systolic and diastolic, rales to the base posterior chest, mild hepatomegaly, femoral pulse present, no edema
- What is the most likely diagnosis?
  - Atrial Septal Defect
  - Atrio-Ventricular Canal
  - Transposition of the great vessels
  - Peripheral pulmonic stenosis
  - Coarctation of the aorta

Summary Slide

Infant sweating while taking a bottle is not normal	It's hard to gain weight if you have a significant heart problem	Not all murmurs are pathological: Stills, Venous Hum
Murmur of an ASD?..VSD?	PDA- Indomethacin to close, Prostaglandin to keep open	Hypertension, absent femoral pulses and rib notching- Coarctation of the Aorta-turner syndrome
Too fast: SVT –vagal maneuvers, adenosine, ECG findings	Too slow: Congential Heart Block- think Maternal Lupus	Fam Hx sudden death, syncope, hearing loss: Prolonged Q-T



THE END

THANK YOU! =)

Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R